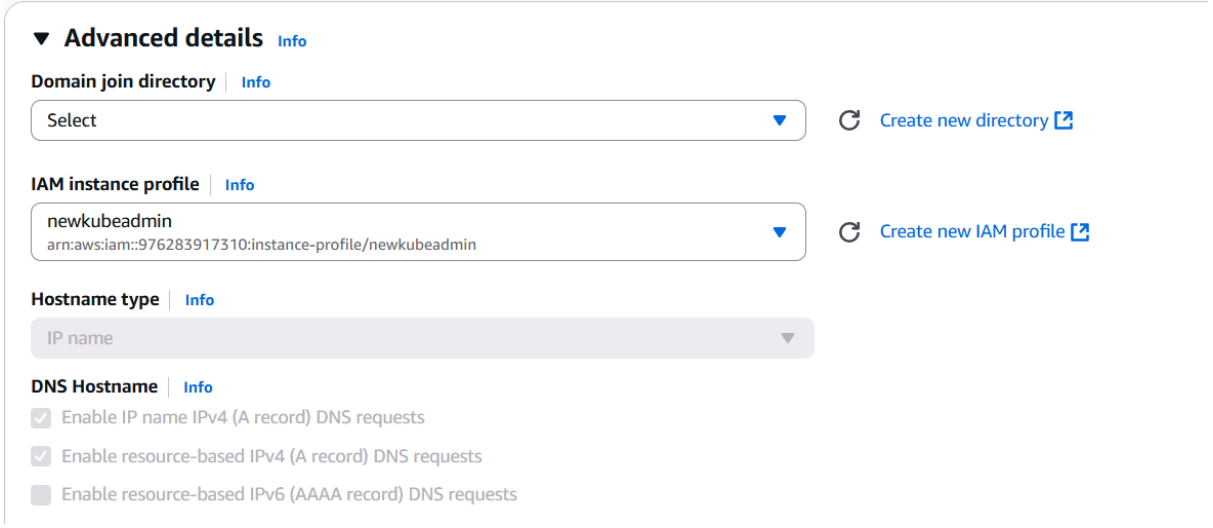
EKS Monitoring setup Document

Go to EC2 Instance -

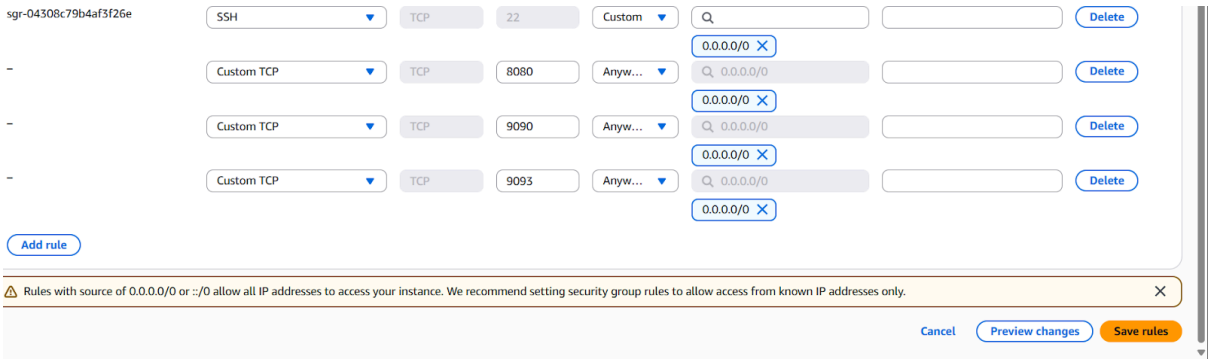
1. **AMI**: Ubuntu 22.04 LTS (x86_64).
2. **Instance type**: t3.small or larger.
3. **Storage**: 20 GB gp3 (or more).
4. **IAM role**: Attach a role with **AdministratorAccess** (or equivalent permissions).
5. **Security group**: Allow inbound **22/tcp** from your public IP. Outbound can remain open.

SSH from your local machine:

CREATE AN IAM ROLE WITH ADMINPERSMISSION SO YOU DONT HAVE TO CONFIGURE AWS MANUALLY



Enable these ports



Open terminal on you local machine where you key is stored:
And paste this after changing your key-pair name and changing public ip
ssh -i "kuberadmin.pem" ubuntu@ec2-184-73-107-7.compute-1.amazonaws.com

```
PS C:\Users\syada\Downloads> ssh -i "kuberadmin.pem" ubuntu@ec2-184-73-107-7.compute-1.amazonaws.com
The authenticity of host 'ec2-184-73-107-7.compute-1.amazonaws.com (184.73.107.7)' can't be established.
ED25519 key fingerprint is SHA256:V88Dv7HHoC6/Q1Ju72IPrAexj4827KLg+5yZ+AG1nrM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-184-73-107-7.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Sep  5 09:34:19 UTC 2025

  System load:  0.25               Temperature:           -273.1 C
  Usage of /:   10.7% of 18.33GB   Processes:             128
  Memory usage: 7%                 Users logged in:       1
  Swap usage:   0%                 IPv4 address for ens5: 172.31.33.44


Expanded Security Maintenance for Applications is not enabled.

19 updates can be applied immediately.
17 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

Update and install basic tools:

sudo apt-get update -y
sudo apt-get install -y unzip curl tar git

2.1 AWS CLI v2

eksctl

curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o awscliv2.zip
unzip awscliv2.zip
sudo ./aws/install
aws --version

If using an IAM role, no configuration needed. Otherwise:

aws configure
Enter Access Key, Secret, region (us-east-1), output (json)

```
ubuntu@ip-172-31-33-44:~$ aws configure
AWS Access Key ID [None]: AKIA6GTYOLP7GXWXHOUW
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
```

2.2 kubectl

KUBECTL_VERSION=$(curl -L -s https://dl.k8s.io/release/stable.txt)
curl -LO "https://dl.k8s.io/release/${KUBECTL_VERSION}/bin/linux/amd64/kubectl"
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
kubectl version --client

```
ubuntu@ip-172-31-33-44:~$ KUBECTL_VERSION=$(curl -L -s https://dl.k8s.io/release/stable.txt)
curl -LO "https://dl.k8s.io/release/${KUBECTL_VERSION}/bin/linux/amd64/kubectl"
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
kubectl version --client
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   138  100   138    0     0   1918      0 --:--:-- --:--:-- --:--:--  1943
100 57.7M  100 57.7M    0     0   219M      0 --:--:-- --:--:-- --:--:--  219M
Client Version: v1.34.0
Kustomize Version: v5.7.1
```

2.3 curl -sL "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.gz" | tar xz -C /tmp

sudo mv /tmp/eksctl /usr/local/bin/
eksctl version

```
ubuntu@ip-172-31-33-44:~$ curl -sL "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.
gz" | tar xz -C /tmp
sudo mv /tmp/eksctl /usr/local/bin/
eksctl version
0.214.0
```

2.4 Helm v3

curl -fsSL https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
helm version

```
ubuntu@ip-172-31-33-44:~$ curl -fsSL https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
helm version
Downloading https://get.helm.sh/helm-v3.18.6-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
version.BuildInfo{Version:"v3.18.6", GitCommit:"b76a950f6835474e0906b96c9ec68a2eff3a6430", GitTreeState:"clean", GoVersi
on:"go1.24.6"}
ubuntu@ip-172-31-33-44:~$
```

3) Create EKS Cluster and Nodegroup

3.1 Create Cluster

eksctl create cluster \
--name observability6 \
--region us-east-1 \
--zones us-east-1a,us-east-1b \
--without-nodegroup

```
2025-09-05 10:09:16 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster,
 eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon i
s via pod identity associations; after addon creation is completed, add all recommended policies to the config file, und
er `addon.PodIdentityAssociations`, and run `eksctl update addon`
2025-09-05 10:09:16 [ℹ]  creating addon: vpc-cni
2025-09-05 10:09:16 [ℹ]  successfully created addon: vpc-cni
2025-09-05 10:11:16 [ℹ]  waiting for the control plane to become ready
2025-09-05 10:11:17 [✔]  saved kubeconfig as "/home/ubuntu/.kube/config"
2025-09-05 10:11:17 [ℹ]  no tasks
2025-09-05 10:11:17 [✔]  all EKS cluster resources for "observability2" have been created
2025-09-05 10:11:18 [ℹ]  kubectl command should work with "/home/ubuntu/.kube/config", try 'kubectl get nodes'
2025-09-05 10:11:18 [✔]  EKS cluster "observability2" in "us-east-1" region is ready
ubuntu@ip-172-31-33-44:~$
```

3.2 Associate OIDC Provider
eksctl utils associate-iam-oidc-provider \

--region us-east-1 \
--cluster observability6 \
--approve

3.3 Add Private Nodegroup

eksctl create nodegroup \
--cluster=observability6 \
--region=us-east-1 \
--name=observability5-ng-private \
--node-type=t3.medium \
--nodes-min=2 \
--nodes-max=3 \
--node-volume-size=20 \
--managed \
--node-private-networking

```
2025-09-05 10:17:51 [ℹ]  checking cluster stack for missing resources
2025-09-05 10:17:51 [ℹ]  cluster stack has all required resources
2025-09-05 10:17:51 [ℹ]  building managed nodegroup stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:17:51 [ℹ]  deploying stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:17:51 [ℹ]  waiting for CloudFormation stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:18:21 [ℹ]  waiting for CloudFormation stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:18:59 [ℹ]  waiting for CloudFormation stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:20:25 [ℹ]  waiting for CloudFormation stack "eksctl-observability2-nodegroup-observability2-ng-private"
2025-09-05 10:20:25 [ℹ]  no tasks
2025-09-05 10:20:25 [✓]  created 0 nodegroup(s) in cluster "observability2"
2025-09-05 10:20:25 [ℹ]  nodegroup "observability2-ng-private" has 2 node(s)
2025-09-05 10:20:25 [ℹ]  node "ip-192-168-72-111.ec2.internal" is ready
2025-09-05 10:20:25 [ℹ]  node "ip-192-168-98-102.ec2.internal" is ready
2025-09-05 10:20:25 [ℹ]  waiting for at least 2 node(s) to become ready in "observability2-ng-private"
2025-09-05 10:20:25 [ℹ]  nodegroup "observability2-ng-private" has 2 node(s)
2025-09-05 10:20:25 [ℹ]  node "ip-192-168-72-111.ec2.internal" is ready
2025-09-05 10:20:25 [ℹ]  node "ip-192-168-98-102.ec2.internal" is ready
2025-09-05 10:20:25 [✓]  created 1 managed nodegroup(s) in cluster "observability2"
2025-09-05 10:20:25 [ℹ]  checking security group configuration for all nodegroups
2025-09-05 10:20:25 [ℹ]  all nodegroups have up-to-date cloudformation templates
ubuntu@ip-172-31-33-44:~$
```

3.4 Update Kubeconfig

aws eks update-kubeconfig --name observability6 --region us-east-1
kubectl get nodes

```
ubuntu@ip-172-31-33-44:~$ aws eks update-kubeconfig --name observability2 --region us-east-1
Added new context arn:aws:eks:us-east-1:976283917310:cluster/observability2 to /home/ubuntu/.kube/config
```

Step 4: Verify Cluster & Nodes

`kubectl get nodes`

```
ubuntu@ip-172-31-33-44:~$ kubectl get nodes
NAME                                       STATUS   ROLES    AGE     VERSION
ip-192-168-72-111.ec2.internal             Ready    <none>   2m44s   v1.32.7-eks-3abbec1
ip-192-168-83-46.ec2.internal              Ready    <none>   4m      v1.32.7-eks-3abbec1
ip-192-168-98-102.ec2.internal             Ready    <none>   2m56s   v1.32.7-eks-3abbec1
ip-192-168-98-73.ec2.internal              Ready    <none>   4m      v1.32.7-eks-3abbec1
ubuntu@ip-172-31-33-44:~$
```

🔹 Step 5: Create Namespace

```
kubectl create ns monitoring
```

🔹 Step 6: Install Monitoring Stack

helm repo add prometheus-community https://prometheus-community.github.io/helm-charts

helm repo update

vim custom_kube_prometheus_stack.yml

> https://raw.githubusercontent.com/akshu20791/k8s-monitoring-using-prometheus/refs/heads/main/custom_kube_prometheus_stack.yml
> here you will find yaml file. Copy the content and put it in yaml file with name `custom_kube_prometheus_stack.yml` in current location where you will run `helm command`

```
helm upgrade --install monitoring
prometheus-community/kube-prometheus-stack \
  -n monitoring \
  -f ./custom_kube_prometheus_stack.yml
```

```
root@ip-172-31-36-221:/home/ubuntu# helm upgrade --install monitoring prometheus-community/kube-prometheus-stack \
  -n monitoring \
  -f ./custom_kube_prometheus_stack.yml
Release "monitoring" does not exist. Installing it now.
NAME: monitoring
LAST DEPLOYED: Fri Oct  3 11:15:56 2025
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
NOTES:
kube-prometheus-stack has been installed. Check its status by running:
  kubectl --namespace monitoring get pods -l "release=monitoring"

Get Grafana 'admin' user password by running:

  kubectl --namespace monitoring get secrets monitoring-grafana -o jsonpath="{.data.admin-password}" | base64 -d ; echo

Access Grafana local instance:

  export POD_NAME=$(kubectl --namespace monitoring get pod -l "app.kubernetes.io/name=grafana,app.kubernetes.io/instance
=monitoring" -oname)
  kubectl --namespace monitoring port-forward $POD_NAME 3000

Visit https://github.com/prometheus-operator/kube-prometheus for instructions on how to create & configure Alertmanager
and Prometheus instances using the Operator.
root@ip-172-31-36-221:/home/ubuntu#
```

🔹 Step 7: Verify Pods

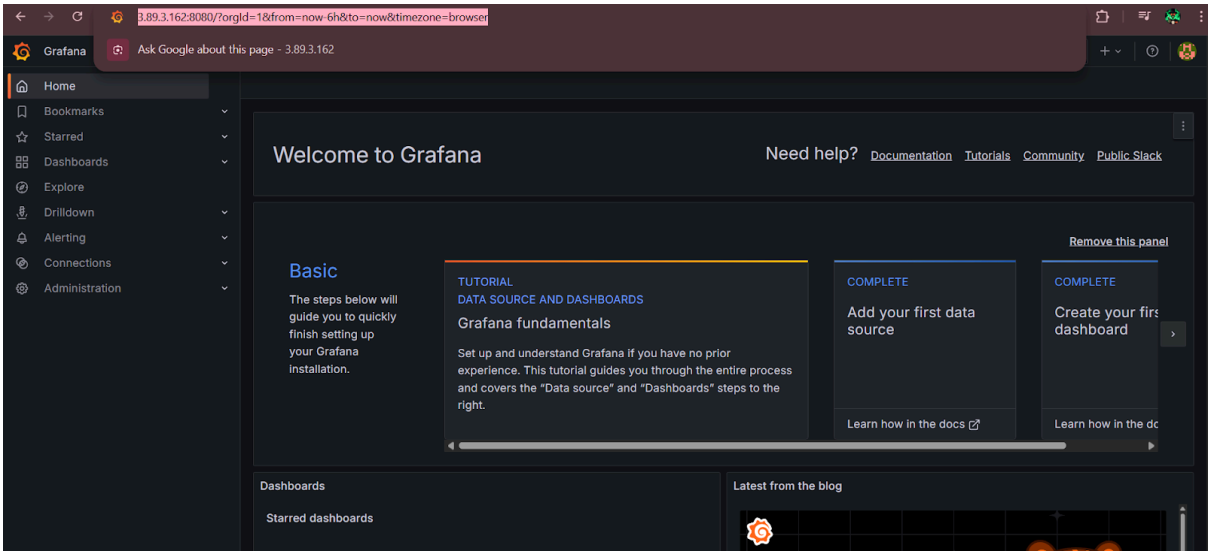
```
kubectl get pods -n monitoring
```

```
root@ip-172-31-33-131:/home/ubuntu# kubectl get pods -n monitoring
NAME                                                     READY   STATUS    RESTARTS   AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running   0          10m
alertmanager-monitoring-kube-prometheus-alertmanager-1   2/2     Running   0          10m
monitoring-grafana-78ccf89d86-k4lpk                      3/3     Running   0          10m
monitoring-kube-prometheus-operator-597c469c59-tkshq     1/1     Running   0          10m
monitoring-kube-state-metrics-7fc7c7469d-62qxf           1/1     Running   0          10m
monitoring-prometheus-node-exporter-lp98f                1/1     Running   0          10m
monitoring-prometheus-node-exporter-z6zd8                1/1     Running   0          10m
prometheus-monitoring-kube-prometheus-prometheus-0       2/2     Running   0          10m
root@ip-172-31-33-131:/home/ubuntu#
```

🔹 Step 8: Access Grafana

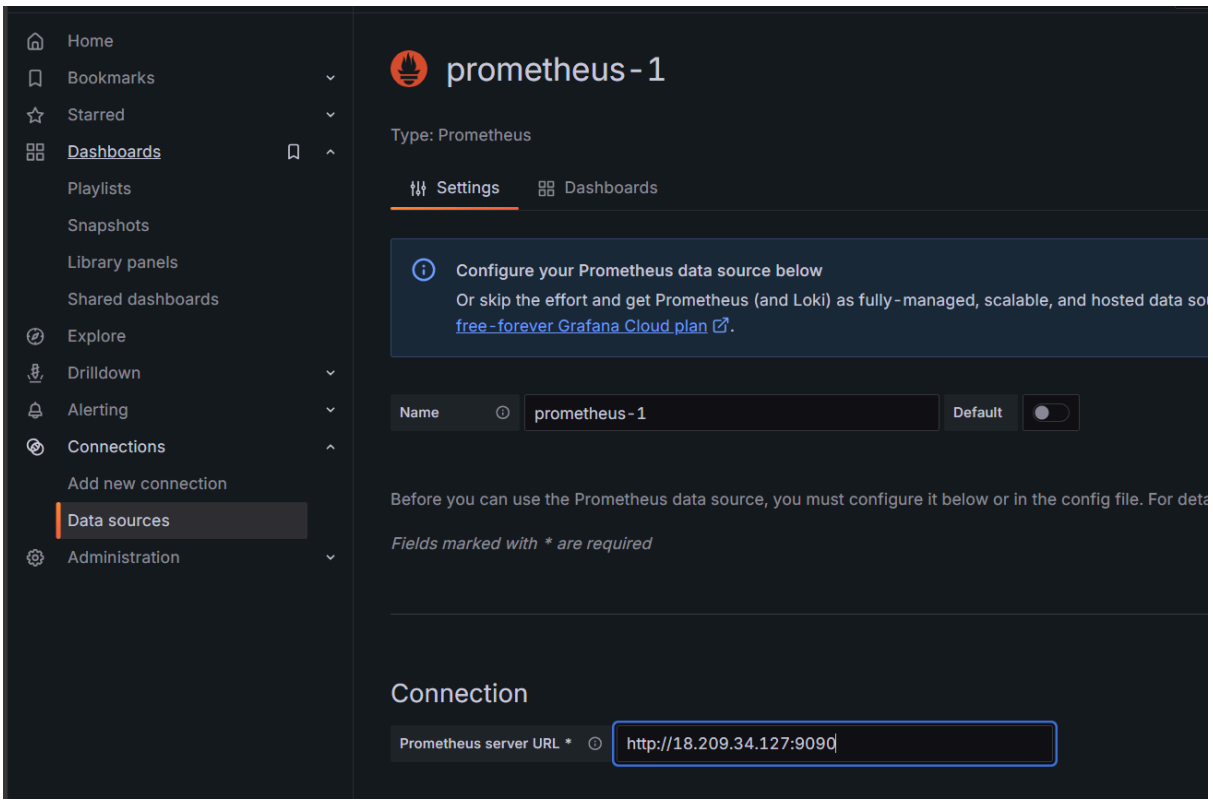
```
kubectl port-forward svc/monitoring-grafana -n monitoring 8080:80
--address 0.0.0.0 &
```

```
(here & is detach mode)
```



Username : admin
Pswrd: prom-operator

Add data source



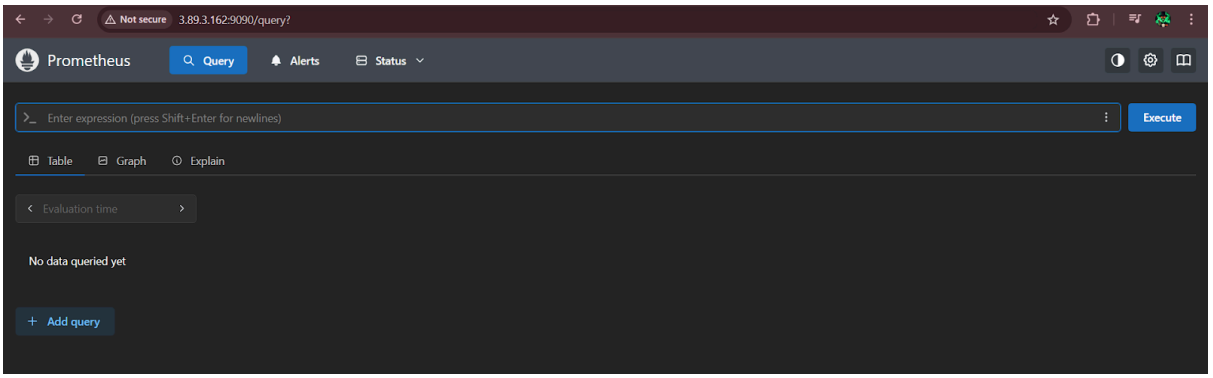
Here this 18.209.34.127 is the ip address of the machine which you created in start (in which you are performing all the task)

Prometheus:

kubectl port-forward service/prometheus-operated -n monitoring 9090:9090 --address 0.0.0.0 &

(here & is detach mode)

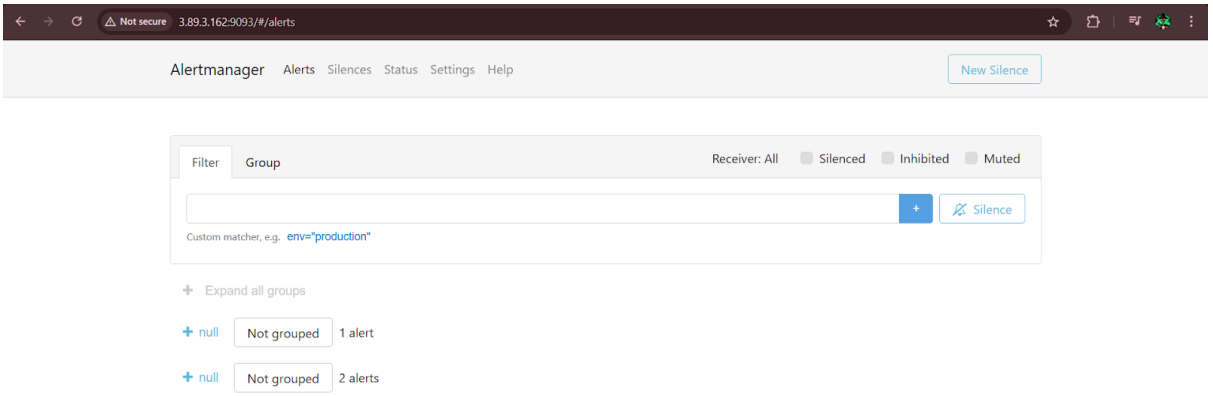
Access: `http://<bastion-public-ip>:9090`



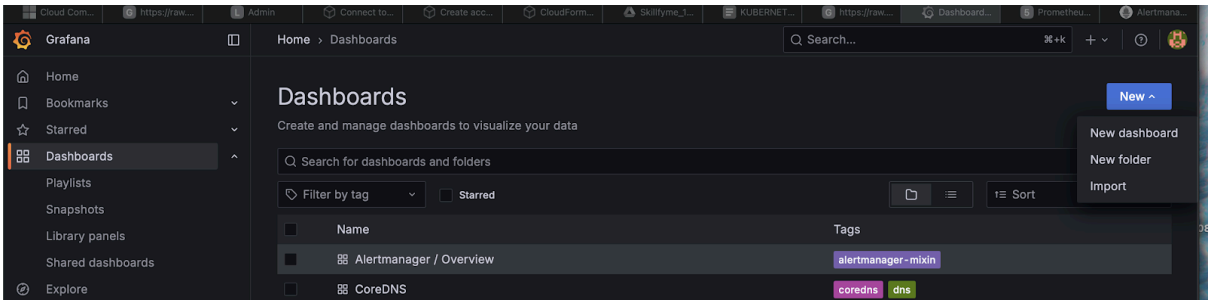
Alertmanager:
kubectl port-forward service/alertmanager-operated -n monitoring 9093:9093 --address 0.0.0.0 &

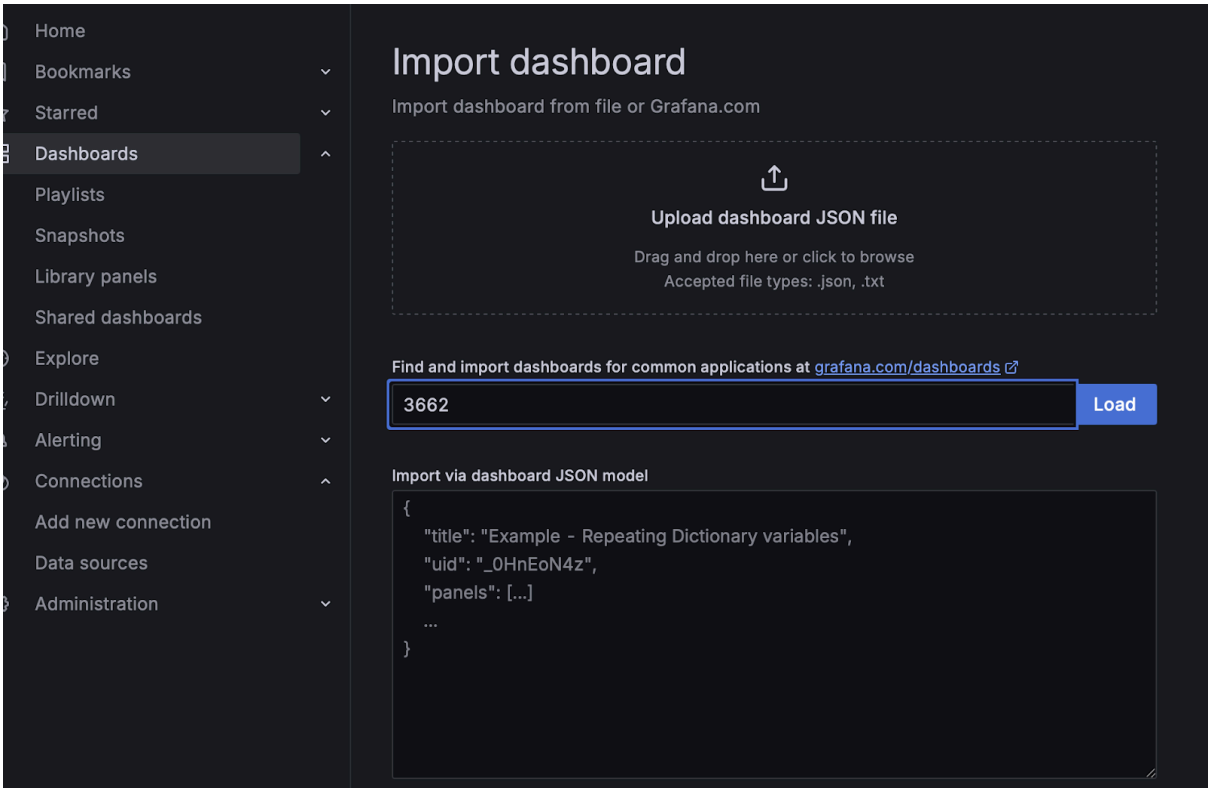
(here & is detach mode)

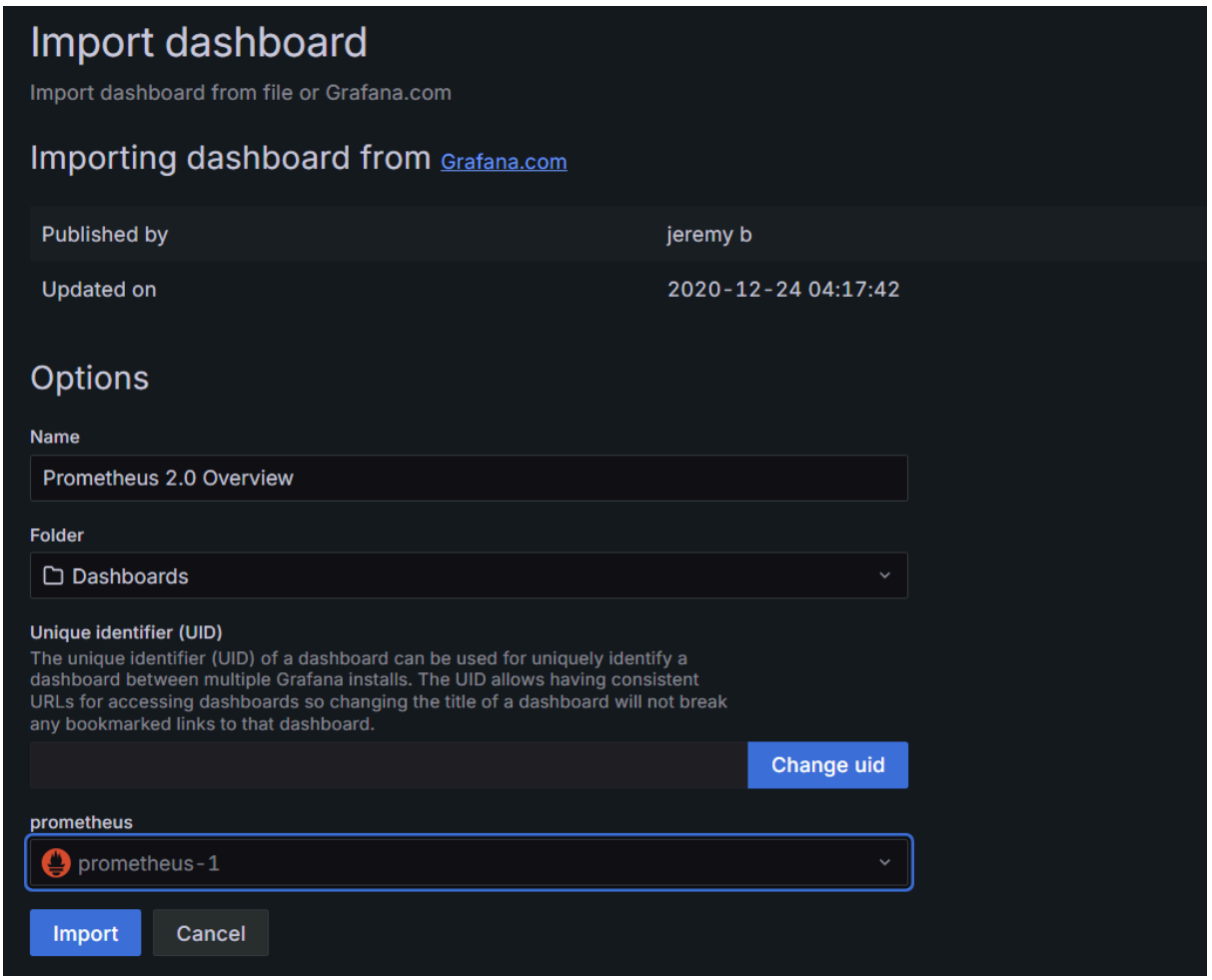
Access: `http://<bastion-public-ip>:9093`



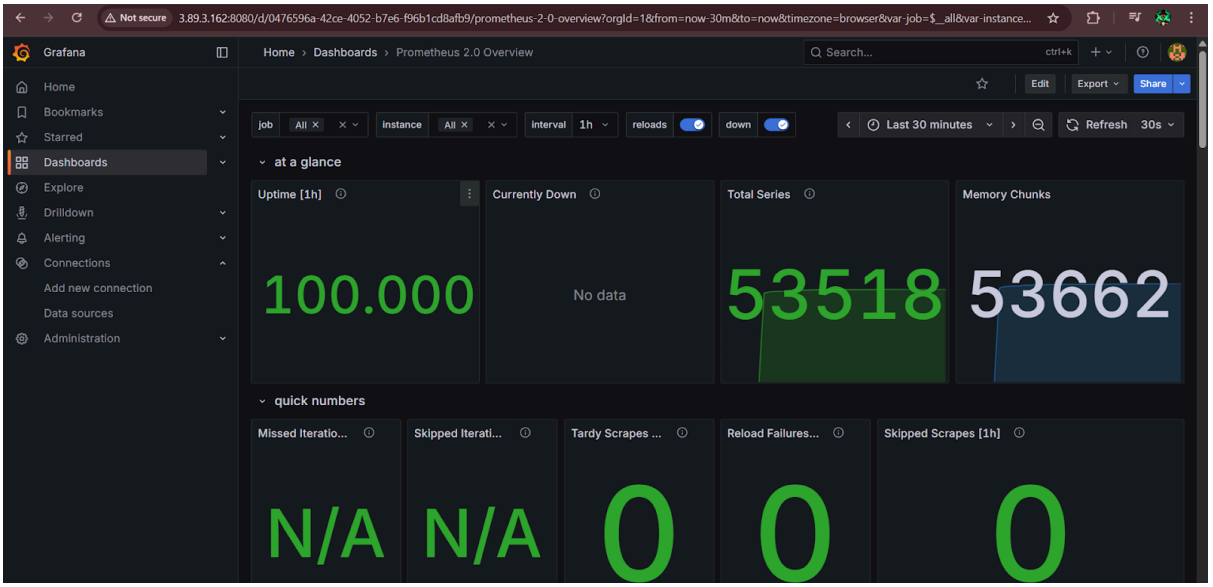
`Import dashboard 3662 or 15757 or 1860`







It will give you the dashboard:



Clean UP

Uninstall helm chart:

```
helm uninstall monitoring --namespace monitoring

Delete namespace:
kubectl delete ns monitoring

Delete Cluster & everything else:
eksctl delete cluster --name observability5
```